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When Front-Stage Innovation Overloads Back-Stage Workers: A Dynamic Model of Workforce, Capability, and Customer Trust in Omnichannel Retail

(Authors' names are not included for peer review)

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Abstract. Problem definition. Digital investment in omnichannel retail generates back-stage demand on workers that standard dashboards cannot see. Front-stage automation amplifies operational complexity faster than back-stage capacity can absorb it, creating a structural measurement gap: technology returns and labour consequences appear on different ledgers, and the workforce that determines whether technology delivers its designed capability goes systematically underinvested.

Methodology/results. We formalise this gap as a parsimonious dynamic model and derive three analytically characterised results. (i) A computable digitalisation ceiling exists beyond which additional front-stage investment destroys customer lifetime value by overloading back-stage workers; front-stage demand metrics continue rising even as lifetime value collapses. (ii) The entire return to workforce investment flows through service capacity, invisible to dashboards that treat payroll as a cost line. (iii) The system is bistable: at our baseline calibration, the neglected firm (lowest workforce investment) faces a safe rollout window approximately 37% the size of its investing peer. Three empirical strategies provide directionally consistent evidence across different identification strategies. A statistically significant inverted-U between e-commerce share and customer satisfaction, and a post-shock decline in the American Customer Satisfaction Index following major omnichannel launches, support the digitalisation ceiling. County-level data from the Bureau of Labor Statistics ($N = 30,951$ county-year observations across 3,115 US counties, 2015–2024) support a labour-market analogue of the Good Jobs Buffer on two margins: counties with higher initial retail wages show significantly more resilient employment ($p = 0.025$) and establishment counts ($p < 0.001$) as e-commerce grows, though the extensive margin has a pre-existing trend that limits causal attribution.

Managerial implications. The model supplies two computable numbers absent from standard dashboards: a digitalisation ceiling and a maximum safe rollout size. Both are workforce-contingent, making workforce investment a structural requirement of digital strategy rather than a social concession.

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Key words: omnichannel retail, workforce well-being, service capability, customer trust, system dynamics, people-centric operations

1. Introduction

Omnichannel grocery retailers have invested heavily in digital capabilities in recent years (Hübner et al. 2021, Bijmolt et al. 2021), enabling customers to build baskets across channels, reserve items for same-day collection, and track substitutions in real time. Yet as digital engagement has grown, retailers have encountered a structural asymmetry in how costs and returns are measured. Technology investment registers as successful in the front-stage domain where it is evaluated, while the back-stage workforce that must fulfil the digitally generated demand absorbs consequences that appear nowhere in the technology business case (Corbett 2024).

The asymmetry is structural, not incidental. Front-stage digital capabilities multiply handoff complexity far faster than order volume grows (Saghiri et al. 2017), placing distinctive demands on back-stage workers. Workforce investment generates returns that are large and invisible to standard metrics (Fisher et al. 2021, Kesavan et al. 2022, Ton 2014), yet the causal chain runs through mechanisms that dashboards do not monitor.

What is missing is a formal account of the mechanism. The service profit chain (Heskett et al. 1994) and the Good Jobs Strategy (Ton 2014) establish that workforce investment pays off; neither predicts a computable threshold at which technology investment reverses sign, nor a cliff beyond which recovery requires structural intervention. These questions require a model of co-evolving state variables responding to a controllable investment sequence (see §2).

This paper formalises the economic structure of this problem as a parsimonious dynamic model. Technology and workforce are not independent inputs to service capability but strategic complements (Milgrom and Roberts 1990, Bainbridge 1983); in the spirit of Kremer (1993), where the weakest link constrains system output, the capacity multiplier formalises the dependence of technology delivery on workforce engagement. The model nests the quality-erosion framework of Oliva and Sterman (2001) as a special case (EC §EC.2). The central modelling innovation is the *workforce capacity multiplier*: effective service capability is the product of the technology stock and a workforce engagement index. When front-stage demand grows faster than back-stage workforce capacity, a service gap opens that depletes workforce well-being, which reduces

effective capability, which widens the gap further: a reinforcing deterioration that single-domain models cannot generate.

Three formally characterised results follow. A *Front-Stage Trap* (Theorem 1): a computable digitalisation threshold exists beyond which additional front-stage investment reduces customer lifetime value by overloading back-stage workers. A *Good Jobs Buffer* (Theorem 2): the return to workforce investment flows through a capacity-level channel that is invisible to dashboards measuring only operational sensing. A *Burnout Cliff* (Theorem 3): the system exhibits bistability, and at baseline calibration, a firm that neglects its workforce ($f=0.01$) faces a safe deployment window approximately 37% the size of one that invests ($f=0.10$). A fourth result, Proposition 4, establishes that the marginal value of workforce investment is increasing in digitalisation level; Corollary 5 (a consequence of Theorems 1 and 2, extended computationally) shows that the all-technology budget allocation is dominated at and near the Trap threshold, and computationally throughout the tested range. Workforce investment is therefore not a social concession but a structural requirement of digital strategy.

2. Related Literature

Three bodies of research motivate this work, each contributing part of the picture but none assembling it. We review each in turn.

Service quality erosion. The foundational insight that service quality can erode endogenously under demand pressure originates with Oliva and Sterman (2001), whose system-dynamics model (Sterman 2000) produces death spirals in which workers cut corners under demand pressure. Their finding that workers are roughly twice as willing to cut corners as to work overtime grounds the overload-asymmetry parameter σ in the present model. Subsequent work extends the logic to capability erosion (Rahmandad and Repenning 2016, Repenning and Sterman 2001), e-commerce growth (Oliva et al. 2003), and regulatory compliance (Martinez-Moyano et al. 2014). KC and Terwiesch (2009) confirm empirically, in a hospital context, that sustained high workload degrades service outcomes—a mechanism this model formalises for retail. Corbett (2024) calls for formal dynamic models of workforce well-being in operations.

Gap 1: no model in this tradition treats workforce well-being as a positive capability multiplier. **Gap 2:** no Oliva–Sterman extension adds customer trust, a front-/back-stage split, or digital scaling.

Workforce investment and retail performance. A substantial empirical literature documents the workforce-performance link. The Good Jobs Strategy (Ton 2014) identifies four operational choices that drive outperformance; causal evidence shows that responsible scheduling raises productivity (Kesavan et al. 2022) and that profit-maximising staffing exceeds prevailing practice (Fisher et al. 2021, Dubé et al. 2020). The conceptual mechanism is the *service profit chain* (Heskett et al. 1994), which has not been formalised as a dynamic model with computable boundary conditions. The Job Demands-Resources (JD-R) model (Bakker and Demerouti 2007, Demerouti et al. 2001) provides the micro-foundation: overload depletes well-being faster than equivalent surplus restores it, and productivity scales with engagement (Mas and Moretti 2009).

Prospect theory (Kahneman and Tversky 1979, Tversky and Kahneman 1992) provides a deeper behavioural foundation: losses loom larger than equivalent gains by a factor of approximately 2.25, and both domains exhibit diminishing sensitivity. Applied to the workforce, overload-induced depletion is experienced more intensely than equivalent recovery—a prediction remarkably consistent with the 2:1 asymmetry in Oliva and Stermann (2001) and the JD-R model. Loss aversion thus independently grounds the parameter $\sigma > 1$ central to Theorem 3 (and strengthens Theorem 1).

Gap 3: no dynamic model formalises the Good Jobs Strategy or the service profit chain with computable boundary conditions.

Omnichannel capability. Digitalisation creates considerable challenges for retail logistics, with fulfilment complexity scaling super-linearly as channel combinations increase (Hübner et al. 2021, Saghir et al. 2017, Bell et al. 2018, Acimovic and Graves 2015). Automation shifts the operator's role toward exception handling (Bainbridge 1983), making the quality of residual human tasks more consequential for overall output (Gans and Goldfarb 2026). Technology–workforce strategic complementarity (Milgrom and Roberts 1990, Aral et al. 2012) motivates the model's central structural assumption that the marginal value of workforce engagement rises with the technology base (§3.1). **Gap 4:** no model formalises front-to-back cross-domain overload. **Gap 5:** the digitalisation-workforce cross-partial has not been derived for services.

Table 1 summarises the five gaps and the model's response to each.

Table 1 Literature gaps and model contributions.

Gap	Closest prior work	This paper
1. No model treats W as a positive capability multiplier	Oliva and Sterman (2001): W as degradation only	Effective capability as the product of technology and workforce engagement
2. No Oliva–Sterman extension adds trust, domain split, or digital scaling	Oliva and Sterman (2001); Martinez-Moyano et al. (2014)	Customer trust as a co-evolving state; front/back domain split; digital scaling
3. Good Jobs Strategy / service profit chain (SPC) unformalised	Ton (2014); Heskett et al. (1994): empirical only	Dynamic formalisation with analytical equilibria and boundary conditions
4. No cross-domain front-to-back overload model	Saghiri et al. (2017): conceptual only	Super-linear back-stage demand as a function of digital depth and customer trust
5. Digitalisation-workforce cross-partial not derived for services	No direct service prior; see also Gans and Goldfarb (2026) (O-ring)	Formally derived: the return to workforce investment rises with the technology base

3. Model

The model has one moving part that prior frameworks lack: effective capability depends on the workforce. The front-stage digitalisation level $F > 0$ is an exogenous managerial control that sets digital feature complexity and thereby determines back-stage demand. The model traces the consequences of choosing F for three co-evolving state variables and, through them, for customer lifetime value (CLV), a standard increasing function of trust calibrated in §5; all theorems require only $\partial \text{CLV} / \partial T^* > 0$. Each of the eight free parameters has a single role with a clean kill condition (Table 2).

3.1. Stocks and Effective Capability

Three state variables evolve jointly. The first, $K(t) > 0$, is the back-stage technology stock: installed infrastructure, warehouse management systems, and process standardisation. The second, $W(t) \in (0, 1)$, indexes workforce well-being—the engagement, energy, and cognitive capacity of pickers, drivers, and click-and-collect associates. The third, $T(t) \in (0, 1)$, measures customer trust: the aggregate confidence that the retailer will deliver reliably. All three are dimensionless and normalised to their natural operating ranges. Throughout, starred variables (K^* , W^* , T^*) denote steady-state equilibrium values; F_0 is the pre-shock digitalisation level; $\Delta F = F - F_0$ is the upgrade step.

The central structural innovation is the *effective capability*. A warehouse management system or substitution-handling process can only deliver its designed performance if the workforce operating it has

the capacity to do so. Effective capability is therefore not the technology stock alone but its product with a workforce engagement multiplier:

$$K_{\text{eff}}(t) = K(t) \cdot h(W(t)),$$

$$h(W) = (1 - p) + pW, \quad p \in (0, 1). \quad (1)$$

Effective capability is the product of the technology stock and a workforce capacity multiplier $h(W)$, ranging from $h(0) = 1 - p$ (zero well-being: only the automation-driven fraction is realised) to $h(1) = 1$ (full design capacity). The cross-partial

$$\frac{\partial^2 K_{\text{eff}}}{\partial K \partial W} = p > 0 \quad (2)$$

is strictly positive for all $p > 0$, establishing technology and workforce as complements in the Milgrom and Roberts (1990) sense.

The parameter p governs complementarity: $p = 0$ decouples workforce from capability; $p = 1$ makes the operation fully labour-intensive. The baseline $p = 0.5$ is an illustrative moderate starting point; estimation is discussed in §7.2. All three theorems hold qualitatively for any $p \in (0, 1)$.

3.2. Demand and the Service Gap

Customer demand on back-stage operations is

$$D(T, F) = (1 + aT) \cdot F^b, \quad (3)$$

where $a > 0$ is the trust-demand coupling and $b > 1$ is the front-stage amplification exponent. The trust-demand coupling reflects the established finding that customer trust drives spending consolidation and order frequency (Sirdeshmukh et al. 2002, McKnight et al. 2002). The super-linearity $b > 1$ captures the well-documented increase in back-stage fulfilment complexity as front-stage channel breadth grows: as shown in §2, Bell et al. (2018) and Acimovic and Graves (2015) document patterns consistent with fulfilment error rates and demand-capacity mismatches rising faster than linearly with channel combinations. We set $b = 1.3$ as an illustrative, structurally conservative baseline motivated by this literature; the Front-Stage Trap requires only $b > 1$, and all three theorems hold qualitatively across $b \in [1.1, 2.0]$ (EC Table EC.3).

The *service gap* measures the shortfall between demand and effective capability:

$$G(t) = D(T(t), F) - K(t) \cdot h(W(t)). \quad (4)$$

When $G > 0$ (overload), customers experience failures. When $G < 0$ (surplus), service runs reliably. We write $[G]^+ = \max(G, 0)$ and $[-G]^+ = \max(-G, 0)$ for the positive and negative parts.

3.3. Dynamics

Three ordinary differential equations translate the overload and surplus mechanisms into rate equations. Under overload, workers cut corners, trust erodes, and capability may spiral downward; under surplus, capacity allows recovery. All rate parameters carry units of months^{-1} .

Capability. The technology stock evolves as

$$\frac{dK}{dt} = r [G]^+ W - \delta K + u, \quad (5)$$

where $r > 0$ is the sensing-adaptation rate, $\delta > 0$ is the depreciation rate, and $u \geq 0$ is direct back-stage investment. The sensing term $r [G]^+ W$ captures organisational learning from service failures (Tucker 2007): the product structure requires both a gap to detect ($[G]^+ > 0$) and workforce capacity to act on it ($W > 0$), reflecting the finding that capability accumulation scales with the workforce knowledge stock (Argote and Epple 1990). Under surplus ($[G]^+ = 0$), the sensing term vanishes and the firm accumulates no capability from successful operations; this asymmetry does not qualitatively alter the results because all three theorems are driven by overload-regime feedback.

Workforce well-being. Well-being rises under surplus capacity and direct investment; it falls through natural decay and overload-amplified depletion:

$$\frac{dW}{dt} = ([-G]^+ + f)(1 - W) - (\mu + \sigma [G]^+) W. \quad (6)$$

Two grouped terms ensure $W(t) \in [0, 1]$. The first drives W upward (vanishing at $W = 1$): capability surplus and direct workforce investment raise well-being with diminishing returns, consistent with the JD-R model (Bakker and Demerouti 2007). The second drives W downward (vanishing at $W = 0$): natural decay (μ) and overload erosion ($\sigma [G]^+$) deplete well-being, with $\sigma > 1$ reflecting the asymmetry documented by Oliva and Sterman (2001) and independently predicted by the loss-aversion ratio (≈ 2.25) from prospect theory (Tversky and Kahneman 1992). All three theorems hold for any $\sigma > 1$ (EC Table EC.3).

Customer trust. Trust builds through reliable service and erodes faster when reliability fails (Sirdeshmukh et al. 2002, Mittal et al. 1998):

$$\frac{dT}{dt} = \beta [-G]^+ (1 - T) - \eta \beta [G]^+ T - \lambda T, \quad (7)$$

where $\beta > 0$ is the trust-update rate, $\eta > 1$ is the erosion asymmetry, and $\lambda > 0$ is natural trust decay. The λT term captures gradual trust erosion absent service experience—customer-base turnover, competitive switching, and memory decay (McKnight et al. 2002)—with a calibrated half-life of approximately 28 months, conservative relative to omnichannel grocery churn rates; all qualitative results hold for any $\lambda > 0$. We set $\eta = 2.0$, below the approximately 2.5:1 ratio reported by Sirdeshmukh et al. (2002).

With all three ODEs specified, we note a technical property. The system (5)–(7) is piecewise-smooth; existence and uniqueness of solutions follow from Filippov's (1988) framework (details and smoothing robustness in EC §EC.1.1 and §EC.1.8).

3.4. State-Space Invariance and Empirical Mapping

The state space $K > 0$, $W \in [0, 1]$, $T \in [0, 1]$ is forward-invariant for all $t \geq 0$; the verification via directional derivatives at each boundary is given in EC §EC.1.1.

Parameter calibration details are in EC §EC.4.

3.5. Feedback-Loop Architecture

Figure 1 shows the causal-loop architecture of the three ODEs, using the conventions of Sterman (2000). The diagram contains four feedback loops, each traceable from the state-variable arrows.

R1 ($T \rightarrow D \rightarrow G \rightarrow T$) has regime-dependent polarity: balancing under surplus, reinforcing under overload (capacity collapses faster than demand falls). **R2** ($G \rightarrow W \rightarrow K_{\text{eff}} \rightarrow G$) and **R3** ($W \rightarrow K_{\text{eff}} \rightarrow G \rightarrow W$) are both reinforcing: the service gap depletes the workforce, reducing effective capability and widening the gap further. R2 drives the Front-Stage Trap; R3 drives the Burnout Cliff. **B1** ($G \rightarrow K \rightarrow K_{\text{eff}} \rightarrow G$) is balancing: sensing investment closes the gap, but only when the workforce has the capacity to act ($r[G]^+ W$ in Eq. 5). Loops R2 and R3, created by the capacity multiplier $h(W)$, are novel to this model and have no counterpart in Oliva and Sterman's (2001) two-loop structure.

Figure 1 Causal-loop architecture. Solid boxes are state variables; dashed boxes are exogenous controls. Arrow signs indicate direction of effect; loop polarity is shown in parentheses. The $(\pm)$ notation for R1 indicates regime-dependent polarity: balancing when $G < 0$ (surplus), reinforcing when $G > 0$ (overload). R2 and R3, created by the capacity multiplier $h(W)$, are novel to this model.

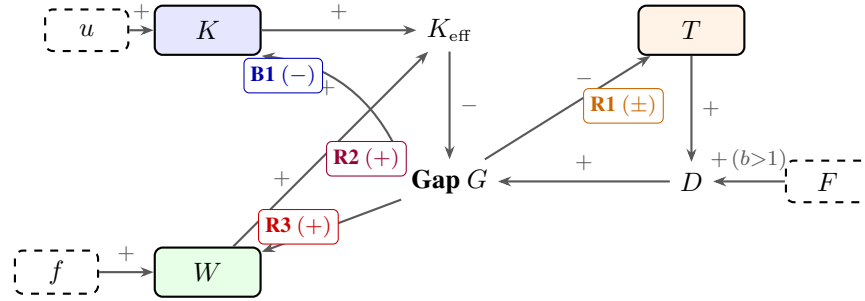


Table 2 Model parameters, baseline values, and kill conditions.

#	Parameter	Baseline	Timescale	Kill	Mechanism killed	Empirical anchor
1	b (demand super-linearity)	1.3	–	$b=1$	Front-Stage Trap	Bell et al. (2018); Acimovic & Graves (2015)
2	a (trust-demand coupling)	0.4	–	$a=0$	Demand feedback	McKnight et al. (2002); Sirdeshmukh et al. (2002)
3	p (workforce complementarity)	0.5	–	$p=0$	Capacity multiplier	Illustrative
4	σ (overload asymmetry)	2.0	–	$\sigma=1$	Tipping point	Oliva and Sterman (2001) [†] ; prospect theory [‡]
5	η (erosion asymmetry)	2.0	–	$\eta=1$	Trust asymmetry	Sirdeshmukh et al. (2002)
6	r (sensing-adaptation)	0.4	$\tau_r \approx 1.7$ mo	$r=0$	Self-correction dies	Tucker (2007); Argote & Epplé (1990)
7	δ (depreciation)	0.03	$\tau_\delta \approx 23$ mo	–	–	Illustrative (EC §EC.4)
8	β (trust-update rate)	0.12	$\tau_\beta \approx 5.8$ mo	$\beta \rightarrow 0$	Trust frozen	Illustrative (EC §EC.4)
+	μ (natural W decay)	0.05	$\tau_\mu \approx 14$ mo	$\mu=0$	W exponential only	Demerouti et al. (2001); Dubé et al. (2020)
+	λ (natural T decay)	0.025	$\tau_\lambda \approx 28$ mo	$\lambda=0$	T exponential only	Illustrative (EC §EC.4)

All time units are months. Timescales are $\ln(2)/\theta$ (half-life). Eight free parameters (rows 1–8); μ and λ govern natural decay. “Illustrative” parameters are calibrated within empirically informed ranges (EC §EC.4). [†]Magnitude comparator from service industry evidence; see §3.3. [‡]Loss-aversion ratio ≈ 2.25 (Tversky and Kahneman 1992) independently predicts $\sigma \approx 2$. Direct estimation of b and p from retailer panel data is the primary empirical extension (§7.2).

3.6. Parameters and Special Cases

Table 2 summarises all parameters. Two control variables complete the model: $u \geq 0$ (capability investment) and $f \geq 0$ (workforce investment). The model recovers the core dynamics of Oliva and Sterman (2001) when $p = 0$ and $a = 0$ (see EC §EC.2 for the structural mapping), and reduces to a Solow-type capital-accumulation equation when $p = 0$, $a = 0$, and $r = 0$.

4. Analytical Results

Four results follow from the model, each identifying a phenomenon that single-domain frameworks cannot produce. Theorems 1–3 characterise three such phenomena; Proposition 4 establishes digitalisation-workforce complementarity. Full proofs are in the Electronic Companion (§EC.1); here we provide proof sketches.

4.1. Equilibrium Structure

The model generates two distinct long-run outcomes. In the first, which we call the *surplus equilibrium*, effective capability exceeds demand. The workforce is healthy, trust builds, and the firm settles into a self-reinforcing high-performance state. In the second, the *low-trust attractor*, demand has overwhelmed capability, the workforce is depleted, and customers have withdrawn trust. Exit from this attractor requires structural intervention: because $r[G]^+W \rightarrow 0$ as $W \rightarrow 0$, the firm cannot learn its way out, and recovery requires direct investment in both u and f .

The formal conditions for each state are as follows. The surplus equilibrium exists for all $F < F_{\text{crit}}$ and is unique under Condition C2 (EC Lemma 2, §EC.1.2). When effective capability exceeds demand, define the operational surplus $S^* \equiv K^*h(W^*) - D(T^*, F) \geq 0$ as the excess of effective capability over demand at equilibrium. Equilibrium trust is an increasing function of S^* , saturating toward one as surplus grows: $T^* = \beta S^* / (\beta S^* + \lambda)$. Equilibrium well-being is similarly increasing in surplus and investment: $W^* = (S^* + f) / (S^* + f + \mu)$. The capability stock settles at $K^* = u/\delta$, determined entirely by the investment-depreciation balance. Full algebraic derivations are in EC §EC.1.2. Both T^* and W^* are increasing in S^* , so any improvement in effective capability propagates into higher trust and well-being at the equilibrium, creating the reinforcing dynamics that underpin the Good Jobs Buffer.

The low-trust attractor is reached whenever demand persistently overwhelms capability. With $T^* \approx 0$, demand collapses to its digitalisation-driven floor and the workforce settles at a low but strictly positive well-being level, held above zero by the workforce investment rate f . Effective capability at this attractor falls short of demand, maintaining a persistent residual gap. The attractor exists whenever $F^b > (u/\delta)h(W_{\text{low}}^*)$ and is locally stable (EC Remark 3).

Maintained conditions. The formal results hold under four regularity conditions (C1)–(C4), verified analytically or numerically across the parameter space (EC Remark 5): surplus equilibrium existence, uniqueness, regularity, and low-trust attractor existence (EC §EC.1.2). The coexistence of surplus equilibrium and low-trust attractor constitutes bistability; the transition at F_{crit} where these two states coalesce is classified as a saddle-node (fold) bifurcation (EC Remark 6).

REMARK 1 (ANALYTICAL TRANSPARENCY). The table below classifies each result by evidentiary basis (Tier A: analytic for all valid parameters; Tier B: analytic under C1–C4; Tier C: computational) and maps it to the empirical evidence in §6.

Result	Tier	Analytical basis	Empirical support (§6)
Thm 1(i): existence	A	IVT + $\Phi(0)$ sign	Inv-U ($p=.028$); event study ($p=.037$)
Thm 1(ii): bounded traj.	A	Gronwall; Lemma 3	–
Thm 1(ii): basin conv.	C	Numerical grid	–
Thm 2: $\partial \text{CLV}^* / \partial f > 0$	B	IFT; GE sign analytic	County empl. ($p=.025$); estab. ($p<.001$) [†]
Thm 2: GE magnitude	C	100 configurations	–
Thm 3(i): $\Delta F_{\text{collapse}}$	A	Direct from Thm 1	Estab. exit ($p<.001$, indirect) [†]
Thm 3(i): convergence	C	Numerical; basin-dep.	–
Prop 4: cross-partial	A	$K_{\text{eff}} = Kh(W)$	Sign (+1.95, $p=.41$; underpowered)
Prop 4: $F_{\text{max}}(W)$	A	Exact at $S^* \rightarrow 0$	–
Cor 5	B/C	IFT / computational	Not tested

[†]Establishment-margin placebo is significant ($p=.012$); see Table 6 and §6.3.

4.2. Theorem 1: The Front-Stage Trap

THEOREM 1 (Front-Stage Trap). Suppose $b > 1$ and $p > 0$. Define the digitalisation threshold

$$F_{\text{crit}} = \left(\frac{u}{\delta} \cdot \left[(1-p) + p \frac{f}{f+\mu} \right] \right)^{1/b}, \quad (8)$$

which depends only on exogenous parameters $(u, \delta, p, f, \mu, b)$ and equals $K^*h(f/(f+\mu))^{1/b}$, the capacity multiplier evaluated at the boundary workforce level. Then:

(i) For $F < F_{\text{crit}}$, the surplus equilibrium (unique under C2; Lemma 2, EC §EC.1.2) exists with $T^* > 0$ and $\text{CLV}^* > \text{CLV}_{\text{low}}^1$ (the CLV at the low-trust attractor).

(ii) For $F \geq F_{\text{crit}}$, no surplus equilibrium exists. All forward trajectories remain in a bounded compact set (Lemma 3, EC §EC.1.2), and the low-trust attractor is locally stable (EC Remark 3). Wide-basin convergence to this attractor is confirmed numerically (Tier C; Remark 1).

The threshold F_{crit} is decreasing in p and increasing in f : the Trap binds earlier when workforce-technology coupling is strong (because a more coupled system depends more heavily on the boundary well-being level $f/(f+\mu) < 1$, reducing effective capacity at the threshold), and is relaxed by workforce investment.

Proof sketch. The surplus equilibrium requires $\Phi(0) > 0$ (where $\Phi(S) \equiv K^*h(W^*(S)) - D(T^*(S), F) - S$ is the surplus residual; full construction in EC §EC.1.2), which holds if and only if $F < F_{\text{crit}}$. At $F = F_{\text{crit}}$, the surplus $S^* = 0$ and the equilibrium ceases to exist. For $F > F_{\text{crit}}$, $G > 0$, activating overload depletion of W through Eq. (6), which reduces $h(W)$, which further widens G through Loop R2. The reinforcing dynamics drive the system toward the low-trust attractor. Boundedness follows from Lemma 3 (EC §EC.1.2): $\dot{K} \leq C(F) - \delta K$ by Gronwall's lemma, and $W, T \in [0, 1]$ by construction. Full proof in EC §EC.1.3. $\square$

At the calibrated baseline, $F_{\text{crit}} \approx 1.11$, corresponding to roughly a 40% increase in digital feature complexity above $F_0 = 0.80$. Figure 2 maps the regime structure: green cells converge to the surplus attractor (high CLV); red cells converge to the low-trust attractor ($\text{CLV} \approx 400$). The cross-domain CLV sign reversal requires $b > 1$ (super-linear demand) and $p > 0$ (workforce-contingent threshold); at $p = 0$ the threshold reduces to $(u/\delta)^{1/b}$ and no longer responds to workforce investment.

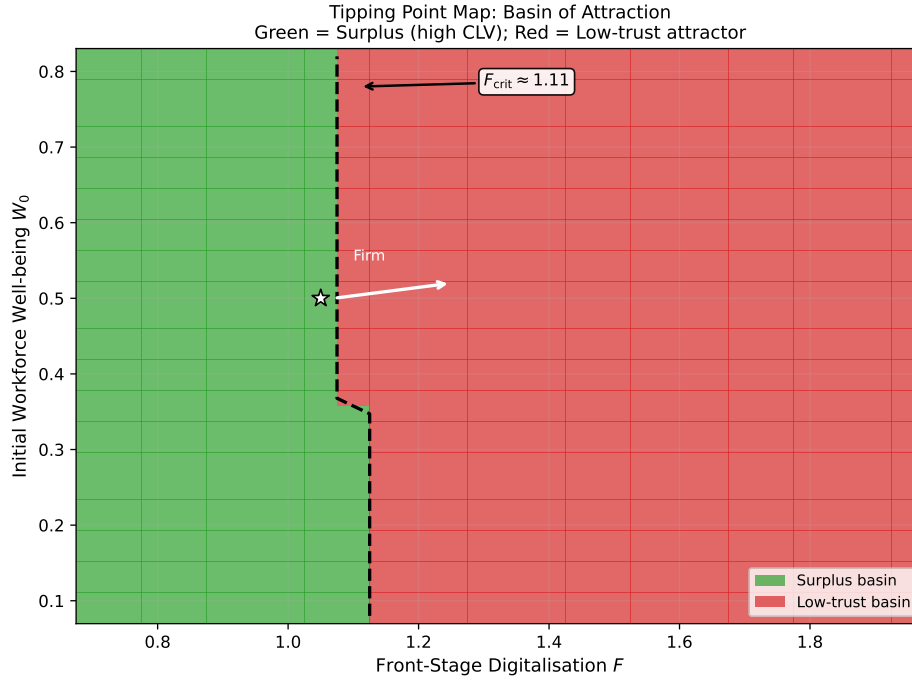
4.3. Theorem 2: The Good Jobs Buffer

THEOREM 2 (Good Jobs Buffer). Suppose $p > 0$ and the surplus equilibrium exists and is stable (Conditions C1–C4, i.e., the surplus equilibrium exists, is unique, and the low-trust attractor is present; Remark 1). Workforce investment f raises steady-state CLV through a multiplicative capacity-level channel:

$$\frac{\partial \text{CLV}^*}{\partial f} > 0 \quad \text{for all } f \geq 0. \quad (9)$$

¹ CLV is formally calibrated in §5; the theorems require only that CLV is increasing in T^* , which holds for any standard customer-lifetime-value functional.

Figure 2 Basin-of-attraction map. Green cells converge to the surplus attractor (high CLV) for $F < F_{\text{crit}} \approx 1.11$; red cells converge to the low-trust attractor for $F \geq F_{\text{crit}}$. Doubled-grid validation confirms boundary stability (EC §EC.4.2).



The return decomposes into a direct capacity effect and an indirect trust effect:

$$\frac{\partial \text{CLV}^*}{\partial f} = \underbrace{\frac{\partial \text{CLV}}{\partial T^*} \cdot \frac{\partial T^*}{\partial S^*} \cdot K^* \cdot p}_{\text{direct: } h(W) \text{ raises } K_{\text{eff}}} \cdot \frac{\partial W^*}{\partial f} + \underbrace{\text{GE trust-demand feedback}}_{\text{indirect: } T^* \rightarrow D, \text{ dampened by } \approx 0.09; \text{ EC §EC.1.4}}.$$

Proof sketch. Since $W^* = (S^* + f)/(S^* + f + \mu)$ at the surplus equilibrium, differentiating gives $\partial W^*/\partial f = \mu/(S^* + f + \mu)^2 > 0$: workforce investment raises steady-state well-being. Higher W^* raises $h(W^*) = (1 - p) + pW^*$, which increases $K_{\text{eff}} = K^* \cdot h(W^*)$, which widens the surplus S^* , which raises $T^* = \beta S^*/(\beta S^* + \lambda)$, which raises CLV. Each link is strictly positive: $\partial h/\partial W = p > 0$, $\partial S/\partial h = K^* > 0$, $\partial T^*/\partial S^* > 0$, $\partial \text{CLV}/\partial T^* > 0$. The chain-rule product is therefore positive, and diminishing returns follow from the concavity of W^* in f . Full proof in EC §EC.1.4. $\square$

The $p=0$ counterfactual in §5.1 confirms this: disabling the workforce multiplier leaves equilibrium CLV exactly invariant to workforce spending. The entire CLV return flows through service capacity, not the operational sensing channel: the indirect sensing channel contributes less than 1% of the total CLV uplift across all tested p values (EC Table EC.4). Standard dashboards, which record technology and workforce returns in separate ledgers, structurally underestimate the financial return to workforce investment.

4.4. Theorem 3: The Burnout Cliff

THEOREM 3 (Burnout Cliff). *Suppose $\sigma > 1$, $p > 0$, and $F_0 < F_{\text{crit}}$ (a surplus equilibrium exists at F_0). Under these conditions the system is bistable in a neighbourhood of F_{crit} (EC Remark 6). Consider a firm at the surplus equilibrium that implements a step increase in F from F_0 to $F_0 + \Delta F$. Two thresholds govern the outcome:*

(i) Permanent-collapse onset. *For $\Delta F \geq \Delta F_{\text{collapse}}$, no surplus equilibrium exists at the post-upgrade level. The low-trust attractor is the only locally stable state (EC Remark 3); convergence to it is confirmed numerically across the studied initial conditions.*

$$\begin{aligned}\Delta F_{\text{collapse}} &= F_{\text{crit}} - F_0 \\ &= \left(\frac{u}{\delta} \left[(1-p) + p \frac{f}{f+\mu} \right] \right)^{1/b} - F_0.\end{aligned}$$

For $\Delta F < \Delta F_{\text{collapse}}$, a new surplus equilibrium exists at the post-upgrade level. Whether a given trajectory reaches this equilibrium depends on its initial condition relative to the basin boundary (Figure 2); numerical verification confirms convergence from the pre-shock equilibrium initial condition across the parameter configurations studied (EC §EC.4.2).

(ii) Instantaneous-overload diagnostic. *A second, larger threshold marks the boundary above which the service gap turns positive at $t = 0^+$:*

$$\Delta F_{\text{rapid}} = \frac{S_0}{b \cdot F_0^{b-1} \cdot (1 + aT^*)}, \quad (10)$$

where $S_0 = K^*h(W^*) - D(T^*, F_0)$ is the pre-upgrade surplus. Because $\Delta F_{\text{rapid}} > \Delta F_{\text{collapse}}$ (at baseline: 0.40 vs. 0.31), a firm in the zone $\Delta F \in [\Delta F_{\text{collapse}}, \Delta F_{\text{rapid}})$ collapses permanently but through a slow trajectory (G negative at $t = 0^+$, surplus eroded gradually). For $\Delta F \geq \Delta F_{\text{rapid}}$, G turns positive immediately and the system enters a rapid self-reinforcing collapse to the low-trust attractor.

The safe zone is $\Delta F < \Delta F_{\text{collapse}}$; ΔF_{rapid} is a deployment-speed diagnostic, not a safety threshold.

Proof sketch. Part (i) follows directly from Theorem 1: for $F_0 + \Delta F \geq F_{\text{crit}}$, no surplus equilibrium exists; for $\Delta F < \Delta F_{\text{collapse}}$, a new surplus equilibrium exists at $F_0 + \Delta F$, and numerical verification confirms that trajectories originating at the pre-shock surplus equilibrium converge to this new surplus equilibrium across the parameter configurations studied (EC §EC.4.2). For Part (ii), setting the first-order demand increment $\Delta D \approx (1 + aT^*)bF_0^{b-1}\Delta F$ equal to the pre-upgrade surplus S_0 gives ΔF_{rapid} . When $\Delta F < \Delta F_{\text{rapid}}$, $G_0^+ = 0$ at t_0^+ and the system crosses the workforce tipping threshold through a slow transient, consistent with the slow-collapse zone. Full proof in EC §EC.1.5. $\square$

Workforce investment f raises F_{crit} and enlarges the safe upgrade zone. Before each deployment phase, the firm should compute $\Delta F_{\text{collapse}} = F_{\text{crit}}(f) - F_0$ as a hard ceiling on planned upgrade size; this ceiling is set entirely by f .

4.5. Proposition 4: Digitalisation-Workforce Complementarity

PROPOSITION 4 (Digitalisation-Workforce Complementarity). *The marginal value of workforce improvement is strictly increasing in the technology base:*

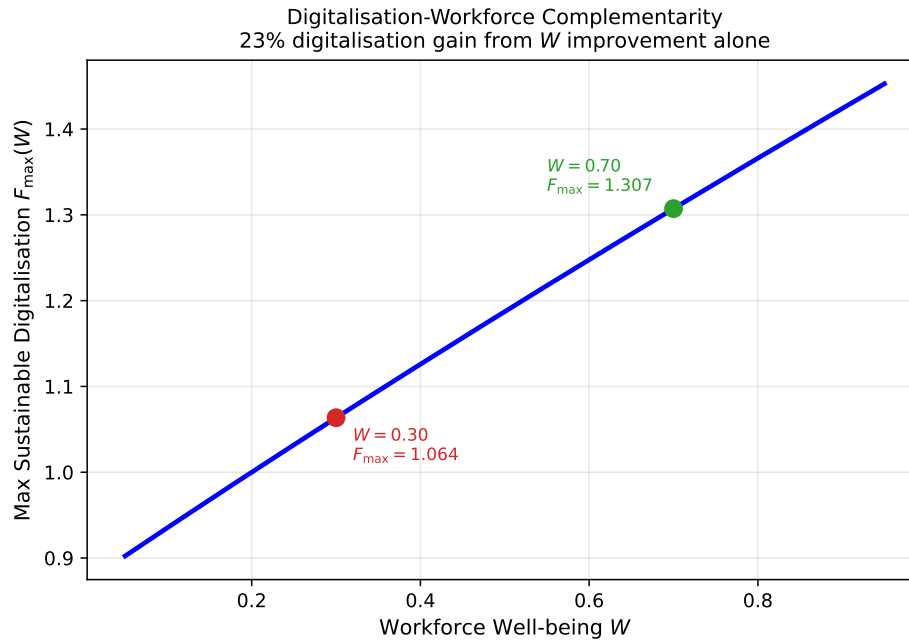
$$\begin{aligned}\frac{\partial K_{\text{eff}}}{\partial W} &= p \cdot K, \\ \frac{\partial^2 K_{\text{eff}}}{\partial W \partial K} &= p > 0.\end{aligned}\tag{11}$$

Since maintaining surplus at higher F requires higher effective capability K_{eff} (that is, either higher K^* through increased u , or higher $h(W^*)$ through increased f), the equilibrium return to workforce investment increases with digitalisation level.

A practical corollary follows: workforce well-being sets a ceiling on how far digitalisation can safely go. At the regime boundary ($S^* \rightarrow 0$), the equilibrium trust approaches zero ($T^* \rightarrow 0$; see Lemma 2, EC §EC.1.2), so:

$$F_{\text{max}}(W) = (K^* \cdot h(W))^{1/b},\tag{12}$$

Figure 3 Digitalisation-workforce complementarity. The healthy firm ($W = 0.70$) sustains 23% higher $F_{\max}$ than the stressed firm ($W = 0.30$), arising entirely from workforce investment.



which is strictly increasing in W for $p > 0$. The full implicit differentiation accounting for $T^*(W, F)$ dependence at interior equilibria (where $S^* > 0$) yields a correction term of order $O(aS^*)$ that vanishes as $S^* \rightarrow 0$; the formula (12) is therefore exact at the boundary and an upper bound on $F_{\max}$ in the interior: trust-elevated demand means true safe capacity is approximately 8% lower than this formula implies at baseline trust levels (EC §EC.1.6).

Proof. The cross-partial follows directly from $K_{\text{eff}} = K[(1-p) + pW]$. The $F_{\max}$ formula identifies F_{crit} at $S^* \rightarrow 0$, where $T^* \rightarrow 0$, giving $(K^*h(W))^{1/b}$, strictly increasing in W since $h'(W) = p > 0$; with the full $T^*(W, F)$ correction this is exact at the boundary and an upper bound in the interior (EC §EC.1.6). The linear $h(W)$ understates complementarity under overload; a concave alternative strengthens the result (EC §EC.1.6). $\square$

Figure 3 illustrates: the gap between a stressed firm ($W = 0.30$, $F_{\max} \approx 1.063$) and a healthy one ($W = 0.70$, $F_{\max} \approx 1.307$) is 23%, arising entirely from workforce investment.

COROLLARY 5 (Budget-Constrained Trap). Under a fixed budget constraint $c_F F + c_f f \leq B$ with $B \geq B_{\min} - \varepsilon$ (where $B_{\min} = c_F \cdot F_{\text{crit}}$), the CLV-maximising allocation includes positive workforce investment

$f^* > 0$: the all- F corner is strictly dominated analytically whenever the budget reaches or exceeds the Trap threshold, and computationally for B throughout $[B_{\min}/2, B_{\min})$ across cost ratios $c_F/c_f \in [0.5, 5]$.

Proof. See EC §EC.1.7. $\square$

5. Computational Study

Numbers sharpen the intuition. We calibrate the model to baseline parameters (Table 2) and simulate three deployment strategies that span the critical threshold. CLV follows Gupta et al. (2004): $CLV = w(T)/(1 - \rho(T))$, where $\rho(T)$ is an annual retention probability (habitual or captive shoppers at the floor) and $w(T)$ an annualised per-customer margin, with $\rho(T) = 0.80 + 0.18T$ and $w(T) = 80(1 + 0.25 \max(T - 0.15, 0))$. The floor $\rho(0) = 0.80$ represents customers who continue purchasing by habit or lack of alternatives (e.g., grocery in underserved areas); $\rho(1) = 0.98$ represents active loyalty. Both endpoints fall within the 70–90% annual range in Gupta et al. (2004); \$80 is an illustrative annual margin for US omnichannel grocery. All analytical results hold for any CLV increasing in trust (calibration in EC §EC.4.3). All three qualitative phenomena are robust across $b \in [1.1, 2.0]$, $\sigma \in (1.0, 3.0]$, and $p \in [0.25, 0.75]$ (EC Table EC.3; at $\sigma = 1$ the Burnout Cliff disappears).

5.1. The Hidden Cost of App Innovation (Figure 4)

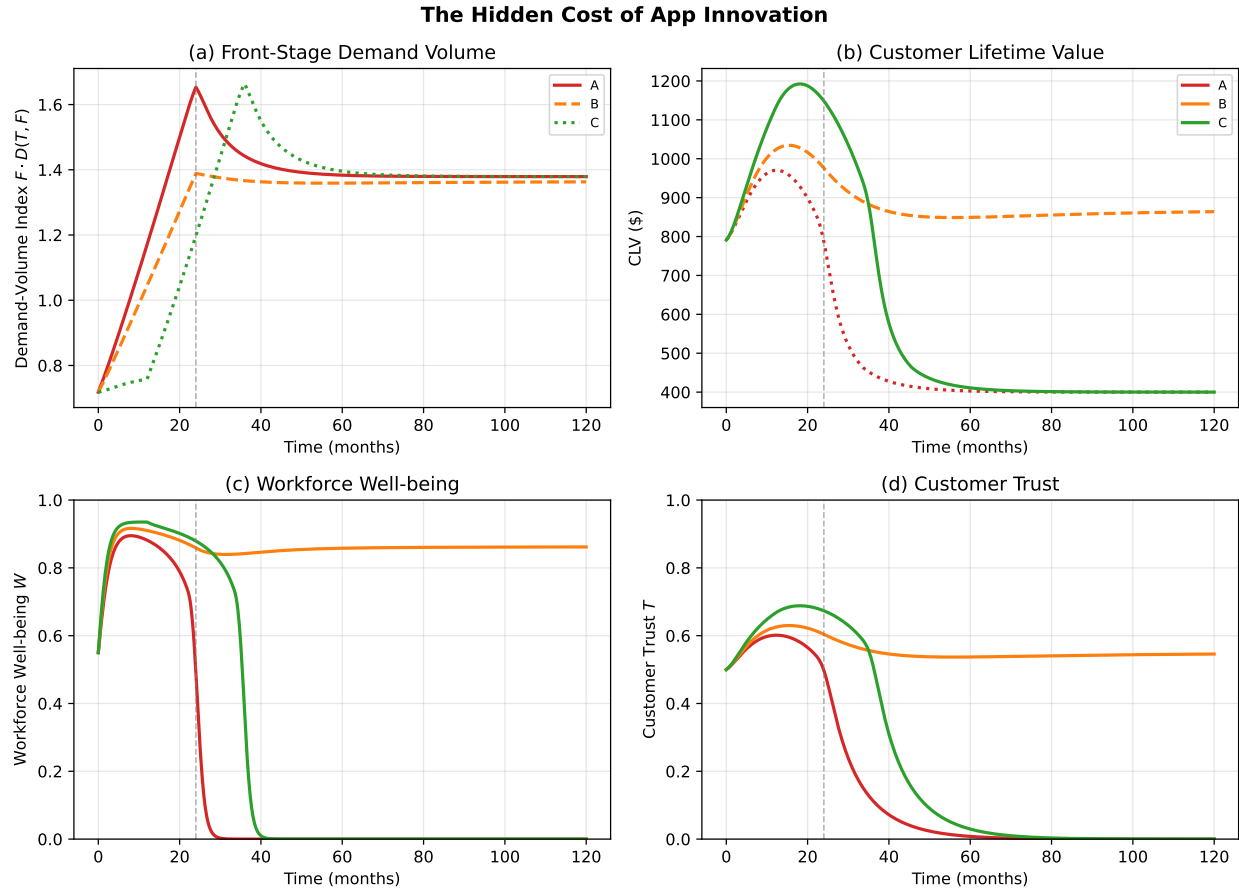
Three deployment strategies span the critical threshold $F_{\text{crit}} \approx 1.11$. All begin from $(K_0, W_0, T_0) = (1.60, 0.55, 0.50)$ with $F_0 = 0.80$. Scenario A ramps aggressively to $F_{\text{end}} = 1.15$ with no workforce investment; Scenario B ramps to $F_{\text{end}} = 1.05$ with concurrent support ($f = 0.06$); Scenario C pre-invests in workforce ($f = 0.10$, months 1–12, after which f returns to 0.0) before ramping to $F_{\text{end}} = 1.15$.

All three appear profitable on front-stage metrics (panel a), yet CLV diverges sharply (panel b): by month 24, Scenario C sustains CLV approximately 40% above Scenario A. Scenario C's persistent advantage—despite f returning to zero after month 12—confirms that well-being is a durable stock. Workforce collapse precedes trust collapse (panels c, d), confirming the invisible cross-domain cost.

5.2. Basin-of-Attraction Structure

Figure 2 maps the basin boundary across the joint space of W_0 and F . The boundary shifts rightward as W_0 increases: a firm near the basin boundary can increase its safety margin by raising workforce investment f , which shifts F_{crit} rightward without touching the technology stock.

Figure 4 The hidden cost of app innovation. Panel (a): the demand-volume index $F \cdot D(T(t), F(t))$ rises for all three scenarios, masking divergent back-stage outcomes. Panel (b): CLV diverges sharply by month 24. Panels (c,d): workforce collapse precedes trust collapse in the no-investment scenario.

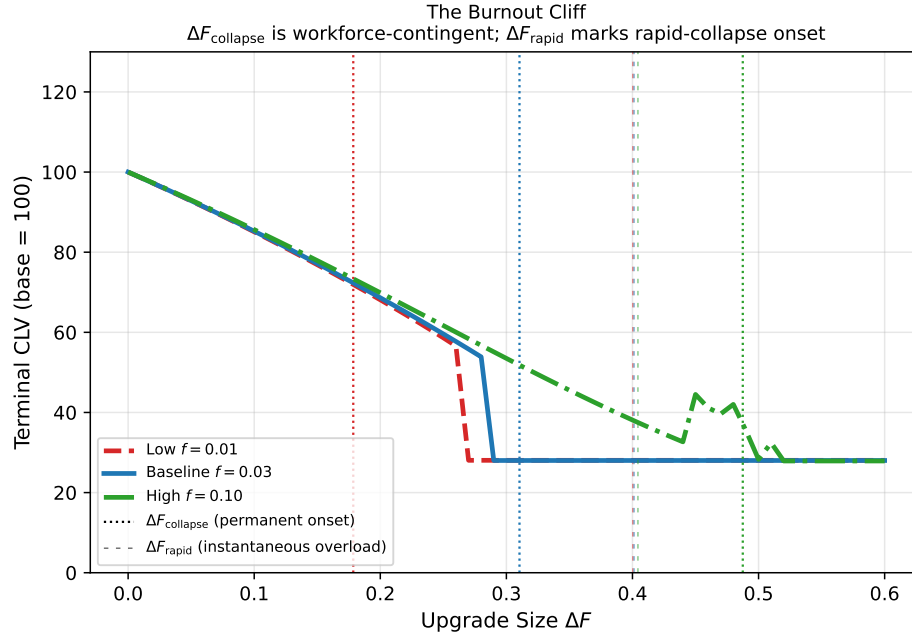


5.3. The Burnout Cliff (Figure 5)

We sweep $\Delta F \in [0, 0.60]$ for three workforce-investment scenarios: neglected ($f = 0.01$), baseline ($f = 0.03$), and invested ($f = 0.10$), each starting from its surplus equilibrium at $F_0 = 0.80$. Varying f shifts $F_{\text{crit}}(f)$ from 0.98 to 1.29, widening $\Delta F_{\text{collapse}}$ from 0.18 to 0.49.

Figure 5 shows the three normalised CLV curves. For the neglected firm ($f = 0.01$), the safe zone collapses to $\Delta F < 0.18$; for the invested firm ($f = 0.10$), it extends to $\Delta F = 0.49$. The slow-collapse zone between $\Delta F_{\text{collapse}}$ and ΔF_{rapid} is the diagnostic blind spot: dashboards report continued growth while trust has already turned negative. For the well-investing firm ($f = 0.10$), the ordering reverses— $\Delta F_{\text{collapse}} = 0.49 > \Delta F_{\text{rapid}} \approx 0.40$ —so the slow-collapse zone disappears entirely (EC §EC.1.5): a well-investing firm faces either a fully safe zone or immediately visible overload. Workforce investment thus purchases perfect

Figure 5 The Burnout Cliff. Normalised terminal CLV versus upgrade size ΔF for three workforce-investment levels. Dotted verticals: $\Delta F_{\text{collapse}}$ (0.18, 0.31, 0.49). The neglected firm ($f = 0.01$) exhausts its safe zone at $\Delta F = 0.18$; the invested firm ($f = 0.10$) retains headroom to $\Delta F = 0.49$.



dashboard visibility of deployment risk, making invisible gradual erosion the exclusive hazard of underinvesting firms.

6. Empirical Evidence

The model makes testable predictions at two levels of aggregation. At the firm level, we test the Front-Stage Trap using an inverted-U specification ($N = 216$, 17 firms) and an event study around major omnichannel launches ($N = 108$, 16 firms²). A cross-sectional complementarity check (Glassdoor $\times$ technology interaction) provides a directional sign check but lacks statistical power ($N_c = 13$). At the county level, we test a labour-market analogue of the Good Jobs Buffer: whether counties with higher initial retail wages show more resilient employment as e-commerce grows ($N = 30,951$ county-year observations, 3,115 US counties). An independent European sample ($N_c = 11$, $N = 20$) is reported in EC §EC.5 as an exploratory check.

6.1. Data and Samples

Table 3 maps each model variable to its empirical proxy.

² Amazon is excluded from the event study because its omnichannel transition was gradual and continuous rather than marked by a discrete launch event, making event-time identification infeasible.

Table 3 Model-to-data mapping.

Model variable	Empirical proxy	Justification
F (digital depth)	E-commerce share (FRED ECOMPCTSA)	National digital penetration index
W (workforce well-being)	Glassdoor overall rating (1–5)	Composite employee engagement
T (customer trust)	ACSI score (0–100)	Aggregate customer confidence
K (technology stock)	CapEx / Revenue	Back-stage capital intensity

Firm-level sample. We assemble an annual panel of 17 publicly listed US retailers from 2008 to 2024 using SEC EDGAR 10-K filings. Customer satisfaction (T proxy) is measured by the American Customer Satisfaction Index (ACSI; Fornell et al. 1996; $N = 216$, 17 firms). Workforce well-being (W proxy) uses Glassdoor overall ratings (1–5 scale; restricted to firm–years with at least five reviews), a publicly available aggregate of employee reviews that has been validated as a measure of employee satisfaction and used as a proxy in employee-wellbeing research (Edmans 2011). Technology investment (K proxy) is capital expenditure as a share of revenue. E-commerce share is from the Federal Reserve (FRED series ECOMPCTSA). The inverted-U specification requires only ACSI and e-commerce ($N = 216$, 17 firms); the complementarity specification additionally requires Glassdoor and CapEx, yielding 131 complete firm–year observations across 13 firms after listwise deletion.

County-level sample. The BLS Quarterly Census of Employment and Wages provides county-level annual average retail employment, weekly wages, and establishment counts for the retail trade sector (NAICS 44–45), 2015–2024. We retain counties with baseline retail employment exceeding 50 and weekly wage exceeding \$200, yielding 3,115 counties and 30,951 county–year observations (199 of the theoretical 31,150 are dropped due to missing QCEW values).

Table 4 reports descriptive statistics for both samples.

6.2. Firm-Level Results

Cross-sectional complementarity (Proposition 4). The between-effects (OLS) estimate of the standardised interaction $K^z \times W^z$ is positive (+1.95, $p = 0.41$; $N = 131$, 13 firms). The sign is consistent with the prediction that firms with jointly higher technology and workforce investment enjoy higher satisfaction, and aligns directionally with Aral et al. (2012). However, the coefficient is not statistically significant, reflecting the extreme power limitation of 13 firm clusters (Cameron et al. 2008). We treat this as a sign check, not

Table 4 Descriptive statistics.

Variable	N	Mean	SD	Range
<i>Panel A: Firm-level sample (13 firms, 2010–2024[‡])</i>				
ACSI score	131	76.5	3.4	68–88
Glassdoor rating (1–5)	131	3.23	0.30	2.7–3.9
CapEx / Revenue (%)	131	2.9 [2.0–4.5] [†]		
<i>Panel B: County-level sample (3,115 counties, 2015–2024)</i>				
Annual avg. retail employment	30,951	5,035	15,458	18–423,900
Annual avg. weekly wage (\$)	30,951	558	151	203–2,861
Baseline wage, 2015 (\$)	3,115	476	110	203–1,697
Baseline employment, 2015	3,115	5,053	15,712	51–416,188
Log employment growth (vs. 2015)	30,951	0.00	0.12	–1.5–2.9
National e-commerce share (%)	10	11.8	3.0	7.4–16.1

[†]CapEx/Revenue reports the median [IQR] rather than the mean and SD due to extreme outliers in the raw ratio (e.g., BBY 2012). Panel A describes the 13-firm subsample with joint ACSI, Glassdoor, and CapEx coverage (used for the complementarity test). [‡]Panel A starts in 2010 because systematic Glassdoor coverage begins circa 2010; earlier observations are excluded by listwise deletion. The inverted-U and event-study specifications use a broader 17-firm and 16-firm ACSI-only sample ($N = 216$ and $N = 108$, respectively); descriptive statistics for these samples are available upon request.

a test. The within-firm TWFE estimate of the interaction is negative and marginally significant (-0.56^* , $p < 0.10$). This does not reflect the structural relationship: within-firm Glassdoor SD is only 30% of total SD, so in years when a firm simultaneously raises CapEx and satisfaction, the interaction captures short-run managerial alignment (common-cause co-movement) rather than structural cross-firm complementarity. The negative main effect of K^z in the TWFE (-0.33^{**}) is consistent with the Front-Stage Trap: within firm, years of higher CapEx intensity are associated with lower satisfaction, reflecting operational disruption preceding capability gains. The between-effects estimand, exploiting persistent cross-firm differences, is the appropriate estimand for the structural question (Milgrom and Roberts 1990).

Inverted-U (Theorem 1). We estimate the quadratic specification

$$ACSI_{it} = \alpha_i + \beta_1 Ecom_t + \beta_2 Ecom_t^2 + \mathbf{X}_{it}\theta + \varepsilon_{it}, \quad (13)$$

where α_i are firm fixed effects, $Ecom_t$ is national e-commerce share, and $\mathbf{X}_{it}$ includes workforce controls where indicated. Year fixed effects are excluded because $Ecom_t$ is a national aggregate identical within year; including them would absorb it entirely. Identification exploits the time-series curvature of the national digitalisation trend, shared across all 17 firms. Firm-specific deployment timing would sharpen identification;

the event study provides this. As a directional check, replacing $Ecom_t$ with a firm-specific digitalization index (cumulative technology adoption percentage; $N = 84$, 6 firms with joint ACSI coverage) yields a negative quadratic ($\hat{\beta}_2 = -0.00061$, $p = 0.039$ under $t(5)$, firm FE; turning point at 51% adoption). The result disappears with year FE ($p = 0.26$), and the extreme power limitation (6 clusters) precludes a definitive test; we treat this as a sign check consistent with firm-level non-monotonicity.

The specification ($N = 216$, 17 firms) yields $\hat{\beta}_2 = -0.028$ ($p = 0.028$ under $t(16)$), consistent with the predicted inverted-U. The estimated turning point (7.9% e-commerce share) lies in the mid-2010s, qualitatively consistent with a digitalisation ceiling, though the correspondence is directional, not quantitative.

The coefficient attenuates when workforce controls are included ($N = 190$; $p = 0.66$). If the model is correct that digitalisation damages satisfaction *through* workforce depletion, then controlling for the mediator (Glassdoor) should absorb the e-commerce effect. A mediation decomposition confirms this pattern: adding Glassdoor as a control reduces the quadratic coefficient by 81% (from -0.026 to -0.005 , $p = 0.70$), and Glassdoor itself is a strong predictor of ACSI ($b = 9.78$, $p < 0.001$). The first-stage relationship (e-commerce $\rightarrow$ Glassdoor) is negative but imprecise ($p = 0.19$), reflecting the low within-firm Glassdoor variation that limits all firm-level workforce tests in this sample. This attenuation pattern is consistent with the workforce channel but does not establish mediation causally; reverse causality (declining satisfaction causing workforce disengagement) cannot be excluded without instruments.

Event study (Theorem 1). We exploit the staggered timing of major omnichannel launches (e.g., Walmart curbside 2017, Target Drive Up 2018, Kroger ClickList 2016) as firm-level technology shocks. Using a $[-3, +3]$ year event window around each firm's launch date t_i^* ($N = 108$, 16 firms), the event-study specification is

$$\begin{aligned} ACSI_{it} = & \alpha_i + \gamma_t + \sum_{\tau=-3}^{+3} \delta_\tau \cdot \mathbf{1}[t - t_i^* = \tau] \\ & + \mathbf{X}_{it}\theta + \varepsilon_{it}, \end{aligned} \quad (14)$$

where α_i are firm fixed effects, γ_t are year fixed effects, $\tau = t - t_i^*$ is event time, $\delta_{-1} = 0$ is the omitted reference period, and $\mathbf{X}_{it}$ includes pre-shock capital expenditure as a control. A pooled post-shock indicator (a single dummy for all $\tau \geq 1$ periods, estimated separately from the event-time specification) yields an

ACSI decline of -0.71 ($p = 0.037$ under $t(15)$). The period $t=0$ is excluded because it captures transition noise, though its event-time coefficient is itself significant (-0.37 , $p = 0.043$; EC Table EC.6). Individual year dummies are each insignificant ($\delta_{+1} = -0.17$, $\delta_{+2} = -0.30$, $\delta_{+3} = -0.90$; EC Table EC.6) but jointly yield -0.71 when constrained equal, illustrating the power gain from pooling. The result is consistent with the Front-Stage Trap. Event-time coefficients for the level specification show near-zero and insignificant pre-trend coefficients at $t = -3$ and $t = -2$ (relative to $t = -1$), supporting the parallel-trends assumption for the basic difference-in-differences (EC Table EC.6). Pre-trend coefficients for the workforce-moderation interaction are similarly insignificant (EC Table EC.6). The workforce moderation ($\text{post} \times \text{pre-Glassdoor}$) is not detectable ($p = 0.79$), consistent with the low within-firm Glassdoor variation rather than absence of the mechanism.

A caveat on staggered treatment timing: recent work (Callaway and Sant'Anna 2021, Sun and Abraham 2021) shows that TWFE estimators with heterogeneous staggered treatment can be biased. Two features limit this concern here: the sample is small ($N_c = 16$ firms) with launches concentrated in 2016–2019, limiting cohort heterogeneity; and the target estimand is the average post-shock effect, which TWFE recovers consistently under modest heterogeneity. With only 16 treated firms, cohort-specific ATTs from Callaway-Sant'Anna or Sun-Abraham would be highly imprecise; we rely on standard TWFE under our maintained assumption of limited treatment-effect heterogeneity across 2016–2019 launch cohorts, supported by the pre-trend evidence.

6.3. County-Level Good Jobs Buffer

The firm-level data cannot deliver a well-powered test of the capacity-multiplier channel. We therefore test an *analogue* of the Good Jobs Buffer at a different margin (employment resilience rather than customer satisfaction) using BLS county-level data, where the dramatically larger sample ($N = 30,951$) provides the statistical power that the firm-level analysis lacks.

Design. The model predicts that workforce investment f buffers the negative effects of digitalisation on service capacity (Theorem 2). The county-level test maps this prediction to an observable analogue: do counties with higher initial retail wages (a proxy for local workforce-investment conditions) show more resilient

retail employment as national e-commerce penetration grows? We test two dependent variables: log employment growth (intensive margin) and log establishment-count growth (extensive margin). Establishment exit is the most observable real-world consequence of permanent capacity collapse (Theorem 3): a retailer that can no longer sustain service quality closes its location.³ Three mapping choices warrant acknowledgement: the outcome is employment and establishment resilience (not CLV); county wages are a noisy proxy for individual-firm f (reflecting local labour-market conditions); and $Ecom_t$ is a national macro trend whereas F is a firm-level choice. Despite these distinctions, the qualitative prediction that higher workforce investment buffers against digitalisation-induced capacity loss is preserved. Initial retail wage is measured in 2015 (before the e-commerce acceleration), standardised cross-sectionally. The interaction $W_{c,2015}^z \times Ecom_t$ captures whether high-wage counties differentially maintain employment and establishments under digital pressure:

$$\begin{aligned} \ln E_{ct} - \ln E_{c,2015} = & \alpha_c + \gamma_t \\ & + \beta_1 (W_{c,2015}^z \times Ecom_t) + \varepsilon_{ct}. \end{aligned} \quad (15)$$

Because $Ecom_t$ is a national-level variable (identical across counties within each year), its main effect is perfectly collinear with the year fixed effects γ_t and is therefore absorbed by them. Likewise, $W_{c,2015}^z$ is time-invariant and absorbed by the county fixed effects α_c . The coefficient β_1 thus identifies the *differential* effect of e-commerce growth on counties with higher initial retail wages, which is the quantity of interest for the Good Jobs Buffer.

Results. The interaction is positive and significant for both margins. On the intensive margin (employment), $\hat{\beta}_1 = 0.0026$ ($t = 2.24$, $p = 0.025$; county and year FE, clustered at county): a county one SD above the mean in initial retail wage ($\approx \$110/\text{week}$) retains 2.6 percentage points more retail employment per 10-point increase in e-commerce share. The long-difference specification (2015 vs. 2023) confirms: high-wage counties gained 2.1 pp more retail employment over the period ($t = 2.57$, $p = 0.010$).

³ State minimum wages do not generate the buffer effect (interaction with e-commerce: $p = 0.22$, clustered at state level). In a specification including both the county-wage and state-MW interactions, the county-wage coefficient strengthens ($p = 0.020$) while the MW coefficient is marginally negative ($p = 0.063$). The buffer thus reflects employer-level wage decisions above the mandated floor, consistent with the model's f as a firm-level choice (EC §EC.5).

On the extensive margin, the buffer is stronger. The establishment-count interaction is $\hat{\beta}_1 = 0.0042$ ($t = 4.64$, $p < 0.001$), 63% larger than the employment coefficient. A county one SD above the mean retains 4.2 percentage points more retail establishments per 10-point increase in e-commerce share. The urban effect is pronounced ($p < 0.001$; 1,557 counties); the rural effect is not significant ($p = 0.17$), mirroring the employment-margin pattern. The decomposition shows that surviving establishments in high-wage counties are somewhat smaller (employment per establishment: $\hat{\beta}_1 = -0.0017$, $p = 0.036$), consistent with the buffer preserving smaller retail businesses that would otherwise exit.

Robustness. For employment (Panel A), the result strengthens when COVID years are excluded ($\hat{\beta}_1 = 0.0031$, $p = 0.014$), holds in urban counties ($p = 0.006$), and is insignificant in rural counties ($p = 0.58$). A pre-period placebo (2015–2017) shows no effect ($p = 0.40$), confirming the buffer activates only under substantial digital pressure. The effect operates through employment quantity, not wages (wage-growth outcome: $p = 0.98$), consistent with the capacity-multiplier channel. For establishments (Panel B), the pattern is similar: robust to COVID exclusion ($p < 0.001$), concentrated in urban counties ($p < 0.001$), and insignificant in rural counties ($p = 0.17$). However, the establishment placebo is significant ($p = 0.012$), indicating that high-wage counties were already gaining establishments relative to low-wage counties before the main e-commerce acceleration. The extensive-margin result therefore reflects a combination of the e-commerce buffer and a secular agglomeration advantage; the employment-margin result, whose placebo is clean, provides the sharper causal test.

A COVID caveat: the e-commerce share jump from 10.7% (2019) to 14.0% (2020) represents forced adoption rather than strategic choice. The result's robustness when COVID years are excluded ($p = 0.014$) confirms the effect is not driven solely by pandemic-era variation.

Tables 5 and 6 present the full regression results.

6.4. Assessment

Three empirical strategies support the model's qualitative predictions. The inverted-U ($p = 0.028$) and the post-shock ACSI decline ($p = 0.037$) both support the Front-Stage Trap (Theorem 1). The county-level wage–e-commerce interaction provides a well-powered test of the Good Jobs Buffer (Theorem 2) on two

Table 5 Firm-level regression results.

	<i>Complementarity</i>		<i>Inverted-U</i>	
	(1) OLS	(2) TWFE	(3) Firm FE	(4) + W
K^z	-0.92 (0.73)	-0.33** (0.13)		
W^z	2.84*** (0.51)	2.00** (0.68)		
$K^z \times W^z$	1.95 (2.36)	-0.56* (0.31)		
Ecom			0.44* (0.23)	0.03 (0.27)
Ecom ²			-0.028** (0.012)	-0.005 (0.012)
Glassdoor				9.78*** (1.80)
Firm FE	No	Yes	Yes	Yes
Year FE	No	Yes	No	No
N	131	131	216	190
Firms	13	13	17	17
R^2_{within}	—	0.44	0.18	0.51

Dependent variable: ACSI score. Cluster-robust standard errors (at firm level) in parentheses; p -values use $t(G-1)$ degrees of freedom. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$. K^z and W^z are standardised CapEx/Revenue and Glassdoor overall, respectively. Ecom is national e-commerce share (%). Columns (1)–(2): complementarity specification ($N=131$, 13 firms). Columns (3)–(4): inverted-U specification ($N=216/190$, 17 firms). Turning point for column (3): 7.9%. Attenuation of Ecom² in column (4) is consistent with the model's prediction that e-commerce damages satisfaction through workforce depletion. The negative within-firm TWFE estimate in column (2) reflects low temporal variation in Glassdoor ratings (within-firm SD is 30% of total SD); see text. Column (1) is OLS without fixed effects; R^2_{within} is not applicable (—); overall R^2 for column (1): 0.55.

Table 6 County-level regression results: Good Jobs Buffer.

	(1) Main	(2) Excl. COVID	(3) Urban	(4) Rural	(5) Placebo
<i>Panel A: Employment growth (intensive margin)</i>					
$W^z_{c,2015} \times \text{Ecom}_t$	0.0026** (0.0011)	0.0031** (0.0013)	0.0050*** (0.0018)	0.0008 (0.0014)	0.0023 (0.0027)
R^2_{within}	0.008	0.013	0.042	-0.000	0.001
<i>Panel B: Establishment growth (extensive margin)[†]</i>					
$W^z_{c,2015} \times \text{Ecom}_t$	0.0042*** (0.0009)	0.0045*** (0.0010)	0.0071*** (0.0014)	0.0018 (0.0013)	0.0044** (0.0018)
R^2_{within}	0.031	0.037	0.089	-0.001	0.005
County FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
N	30,951	24,768	15,532	15,419	9,321
Counties	3,115	3,115	1,557	1,558	3,115

Dependent variables: Panel A uses $\ln E_{ct} - \ln E_{c,2015}$ (log employment growth); Panel B uses $\ln \text{Estab}_{ct} - \ln \text{Estab}_{c,2015}$ (log establishment-count growth). Both are relative to 2015. Standard errors clustered at county level in parentheses. *** $p < 0.01$; ** $p < 0.05$. Ecom _{t} main effect absorbed by year FE; $W^z_{c,2015}$ absorbed by county FE. Column (2): excludes 2020–2021. Columns (3)–(4): split at median baseline employment. Column (5): 2015–2017 pre-period placebo.

[†]Panel B placebo is significant ($p = 0.012$), indicating a pre-existing trend in establishment counts favouring high-wage counties; the extensive-margin coefficient should be interpreted with this caveat (see text).

margins: employment ($p = 0.025$) and establishment counts ($p < 0.001$, $N = 30,951$). The low within- R^2 in Panel A (0.008) is expected for a single interaction net of 3,115 county and 10 year fixed effects, comparable to Chetty et al. (2009), and rises to 0.031 for establishments. The establishment-margin coefficient is 63% larger, though the significant pre-trend in the placebo ($p = 0.012$) means the extensive-margin result partly reflects secular agglomeration; the employment margin, whose placebo is clean, is the sharper causal test. The complementarity sign (+1.95, $p = 0.41$) is directionally consistent with Proposition 4 but underpowered. The Burnout Cliff (Theorem 3) and Budget-Constrained Trap (Corollary 5) are validated computationally; the county establishment pattern is directionally consistent with permanent collapse but cannot distinguish bistability from secular decline. A direct test of the Burnout Cliff using store-level deployment data is the most important empirical gap remaining. The contribution is directionally consistent evidence across different identification strategies, levels, and outcomes, subject to §7.2.

7. Discussion

7.1. Contributions and Managerial Implications

The three theorems together provide the formal micro-foundation that connects Ton's (2014) four operational choices to each other through the dynamics of K_{eff} . Each mapping is analogical—Ton's prescriptions address firm-level operational choices whereas the model's parameters index structural mechanisms—but the correspondences are informative. “Operate with slack” corresponds to positive f ; Theorem 2 quantifies the CLV return. “Standardise and empower” corresponds to lower σ ; well-designed processes enlarge the safe zone of Theorem 3. “Offer less” maps onto keeping F below F_{crit} . “Cross-train” maps onto higher p : cross-trained workers raise $h(W)$ at any engagement level (Proposition 4). The model extends Oliva–Sternman with customer trust, a front-/back-stage domain split, and digital scaling: three extensions absent from the quality-erosion framework, now formalised in a single model that, to the best of our knowledge, provides the first dynamic formalisation of the service profit chain (Heskett et al. 1994, 1997) and Good Jobs Strategy (Ton 2014) with computable boundary conditions.

Managerial implications. Four implications follow. First, the maximum safe rollout speed is a workforce constraint (Theorem 3): the firm should verify that $\Delta F_{\text{collapse}} = F_{\text{crit}}(f) - F_0$ exceeds the planned upgrade

size. Second, workforce well-being is a leading indicator: W reaches crisis levels 6–21 months before trust does (EC Table EC.4). Third, the more digital the firm becomes, the more it should invest in its people (Proposition 4): conventional ROI calculations underestimate returns because the dominant capacity-multiplier channel is invisible in standard dashboards (Theorem 2). Fourth, as a direct consequence of Theorems 1 and 2, the all-technology allocation is dominated at and near the Trap threshold (analytically) and throughout the tested budget range (computationally): reallocating the marginal dollar to workforce spending yields a strict CLV gain (Corollary 5).

7.2. Limitations and Directions for Future Research

Model limitations. F is exogenous; the HJB formulation for optimal deployment is non-trivial because the switching manifold $\{G=0\}$ is endogenous (Filippov 1988). Endogenising F would likely strengthen the Trap: a firm that observes workforce distress and slows its rollout avoids collapse, reinforcing the prediction that workforce monitoring is a precondition for safe deployment. Demand grows without bound in F (no market saturation); predictions are most reliable near F_{crit} . Omitting a direct $W \rightarrow D$ channel and using linear $h(W)$ both make results conservative; a concave $h(W) = (1-p) + pW^\gamma$ ($\gamma < 1$), motivated by prospect theory's diminishing sensitivity, would strengthen them.

Empirical limitations. The firm-level panel comprises publicly traded retailers that survived to 2024; firms that plausibly experienced the most severe capacity collapses (e.g., Sears, Toys R Us) exited the sample through bankruptcy, creating survivorship bias that works against detecting the Burnout Cliff in firm-level data. With $N_c = 13$ firm clusters, firm-level power is limited (Cameron et al. 2008); county-level power ($N_c = 3,115$) does not share this limitation. Glassdoor ratings are voluntary and subject to self-selection; the low within-firm variation (SD 30% of total) further limits the interaction's within-firm identification. The county interaction cannot fully separate the Good Jobs Buffer from non-tradeable-retail embedding; more broadly, high-wage counties may be more resilient to any economic shock, not specifically e-commerce, so the county design identifies the differential effect of wages under digital pressure but cannot rule out a general-prosperity channel. The establishment-margin pre-trend (Table 6, Panel B placebo) reinforces this concern. Store-level data would resolve both. The empirical section reports multiple specifications across

three identification strategies; primary hypotheses (Theorems 1–2) were derived from the model before estimation, and all specifications are reported to permit reader assessment.

Extensions. The most important empirical follow-up is store-level identification of p using exogenous technology rollout events (e.g., staggered self-checkout installation), which would test the capacity multiplier directly rather than through county-level analogues. On the theory side, a competitive multi-firm extension with workforce poaching would endogenise the labour market that the present model treats as a background condition, and stochastic demand forcing would convert the deterministic Burnout Cliff into a probabilistic failure-rate surface.

8. Conclusion

Standard operational dashboards measure the front-stage domain that invests in technology, not the back-stage workforce that fulfils the resulting demand and absorbs the consequences. This disconnect is not merely a measurement problem; it is a structural feature of omnichannel operations that makes cross-domain value destruction structurally predictable from two observable managerial choices: the workforce investment rate f and the digitalisation level F . This paper provides the formal instrument for both.

Together, the three theorems establish that the relationship between digitalisation and customer value is non-monotonic (the Front-Stage Trap imposes a computable ceiling), workforce-contingent (the Good Jobs Buffer channels the entire return through service capacity), and bistable (the Burnout Cliff creates a deployment threshold that workforce investment widens by a factor of 2.7 at baseline). These three properties are invisible to dashboards that partition technology costs and workforce costs onto separate ledgers, which is precisely why the structural gap persists.

These results give operations teams two computable numbers standard dashboards do not report: a digitalisation ceiling (F_{crit}) and a maximum safe rollout size ($\Delta F_{\text{collapse}}$). Three empirical strategies provide directionally consistent evidence across different levels of analysis and identification strategies: an inverted-U between e-commerce and ACSI, a post-shock satisfaction decline following omnichannel launches, and county-level evidence that higher retail wages buffer both employment ($p = 0.025$) and establishment counts ($p < 0.001$; the latter with a pre-trend caveat) against e-commerce growth ($N = 30,951$ county–year observations, 3,115 counties). Store-level identification of p using exogenous technology rollout events remains the most important empirical follow-up.

The broader implication extends beyond omnichannel retail: wherever automated front-stage processes generate demand on human back-stage operations, workforce investment is an operational prerequisite for digital strategy, not an optional cost line. The capacity-multiplier structure is plausibly applicable wherever technology adoption generates hidden back-stage load—healthcare, financial services, logistics—though whether the quantitative thresholds apply in these domains requires domain-specific calibration and is a direction for future work.

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